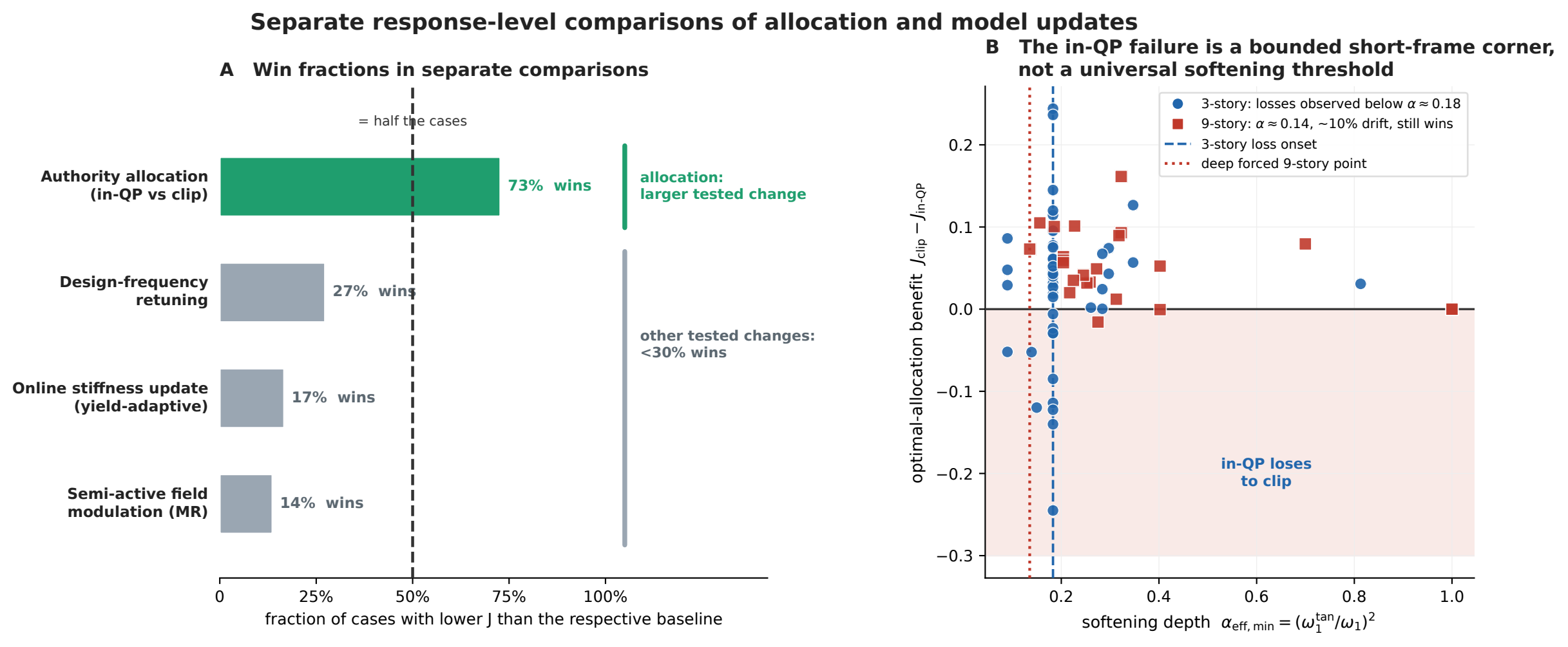

Separate response-level comparisons of allocation and model updates
A Win fractions in separate comparisons
= half the cases
Authority allocation (in-QP vs clip)
73% wins
allocation: larger tested change
Design-frequency retuning
27% wins
other tested changes: <30% wins
Online stiffness update (yield-adaptive)
17% wins
Semi-active field modulation (MR)
14% wins
0
25%
50%
75%
100%
fraction of cases with lower J than the respective baseline
B The in-QP failure is a bounded short-frame corner, not a universal softening threshold
3-story: losses observed below α ≈ 0.18
9-story: α ≈ 0.14, ~10% drift, still wins
3-story loss onset
deep forced 9-story point
optimal-allocation benefit $J_{clip} - J_{in\text{-}QP}$
0.2
0.1
0.0
−0.1
−0.2
−0.3
in-QP loses to clip
0.2
0.4
0.6
0.8
1.0
softening depth $\alpha_{eff,\,min} = (\omega_1^{tan}/\omega_1)^2$